Phase 2 Interim Report

Agriculture information system for Andaman and Nicobar Islands | Run 9e1171369c9e | 16 October 2026

What we made

We wrote a Python program that reads the 17 agriculture files supplied in Phase 1 and saves text and selected numbers in a local SQLite database. Each result keeps its source file and page or table position where available.

The manifest sends each file to a small reader using fixed Python rules. There is no LLM supervisor. The source set includes digital PDFs, an image, maps and an atlas chart, so different readers are used for different formats.

The workflow is a simple sequence: read the manifest, choose a reader, run it, and save the result, status and trace. A failed source is recorded; the code does not retry automatically. For this known list of files, fixed rules are easier to explain than asking a supervisor model to choose.

Readers and source groups

- **PDF table reader:** crop statistics, prices, area and production reports, and the A&N land-cover data table. It preserves table cells and normalizes selected rows.
- **PDF text reader:** island profiles, food-processing report and subsidy report. It saves page text and selected evidence lines.
- **Image OCR reader:** Tesseract reads the census screenshot and eight Total-row values.
- **Map legend reader:** reads map titles, labels, legend items and source notes. It does not create GIS boundaries.
- **Chart reader:** measures five colored bars for Andaman & Nicobar in the LULC atlas. The results are approximate.
- **Hindi label decoder:** a fixed Kruti Dev mapping handles selected labels. This is not machine translation.

Latest local run

Measure	Result
Sources tried / routed	17 / 17
Final statuses	16 loaded, 1 partial; none pending, skipped or failed
Table cells / structured observations	45,139 / 21,158
Non-chart source matches	21,153 / 21,153
Checked crop values / Hindi labels	4/4 / 5/5
Local route, tool and input checks	17/17 each
Automation / unit tests / database	16/17 (94.1%) / 5 passed / integrity ok

Sixteen of seventeen files completed their assigned reader without manual intervention. This is not a claim that every page was turned into a clean database table. Source selection and downloading happen before the run.

One file example

Census screenshot: the manifest identifies an English image, the router chooses *image_ocr_agent*, and Tesseract reads the screenshot plus eight small cells in its Total row. Python stores each recognized value with the source image, page, cell position, reader and run ID. The OCR evidence and tool call are also logged.

Chart example and the main challenge

The atlas chart is on PDF page 102 (printed page 81), Chart 34, for 2022-23. It has colored bars but no printed values. The chart reader compares bar heights with the printed axis and records approximate readings: about 5.0 lakh ha for evergreen/semi-evergreen woodland, 1.7 for degraded woodland, 0.7 for littoral/swamp/mangroves, 0.4 for deciduous woodland and 0.3 for double/triple/annual crop. These are estimates. The separate Phase 1 A&N LULC table is the source for exact figures.

The sources need different tools. pdfplumber reads PDF text and tables; Tesseract 5.5.3 with English OCR reads the screenshot and map labels; a fixed pixel-height rule reads selected chart bars. We tested them on the supplied sources. The maps return labels and legends, not digital geography. We do not claim that OCR or research tools cannot do more; the limits described here are those of the source formats and this implementation.

Hindi and model training

The Basic Statistics PDFs include Hindi labels encoded with the older Kruti Dev font. We use a small fixed character map and keep the original text. Five examples passed; four checked crop values also matched the held-out 2024-25 source. This is not a trained Hindi model. No LLM, adapter or fine-tuning was used, so a base-model versus adapter comparison does not apply unless the instructor requires it as a separate rule.

Database and provenance

SQLite stores the source list, runs, page text, raw table cells, selected observations, text evidence and run traces. Observations include a metric, value, unit, place/time and source coordinates when available. Empty values are not changed to zero. The provenance view joins a result back to the source.

Example: Paddy production is stored as 13,490.2 MT from *basic statistics report 2024-25.pdf*, page 1, table 1, row 4, column 6. The record names the table reader and its run.

Evaluation and LangSmith

The local checks found all source paths and routes. The selected source values matched their saved cells, page text or OCR cells; chart estimates are counted separately. Five unit tests passed and SQLite integrity returned ok.

By reader, tool inputs and expected trajectories matched for image OCR (1/1), chart (1/1), map OCR (2/2), PDF tables (9/9) and PDF text (4/4). These are route and trace checks, not hand checks of every extracted value. Direct checked references are smaller: four numeric values and five Hindi labels.

The latest LangSmith dashboard screenshot shows dataset agriculture-phase2-9e1171369c9e with 17/17 runs and a strict trajectory match average of 1.00. This means the recorded tool sequence matched the expected route; it is not an extraction-accuracy score. The screenshot is included with the run logs.

We used checks for source/agent choice, tool inputs, tool sequence, completion status, source grounding and selected reference values. AgentEvals checked trajectories and LangSmith recorded the 17 examples. We skipped LLM answer-quality, safety and relevance scores because this pipeline does not create conversational LLM answers. Latency and token counts do not help judge these deterministic readers.

What remains limited

- The atlas chart readings are approximate because its bars have no printed numbers.
- The maps have OCR text and legend capture only; there are no GIS boundaries or pixel-based area records.
- Some large PDF tables remain raw cells rather than fully normalized data.
- Source selection, downloads and preparation of checked reference values happen before the automatic run.
- Failures are logged but not retried automatically; this run recorded zero retries and zero manual corrections during processing.

Phase 2 status: The code, corpus, manifest, database, local evaluation, logs, report and demo guide are assembled. One chart remains partial, and the maps are not GIS data. The supplied copies identify government or official publishers, but do not state a reuse licence for every file; a specific GRL is therefore not claimed. This is a working Phase 2 pipeline, not a complete agriculture knowledge base.